

Safety Data Sheet

Dry Vitamin A-Acetate 325 GFP

Revision date : 2018/08/16

Version: 3.1

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(30056973/SDS_GEN_US/EN)

1. Identification

Product identifier used on the label

Dry Vitamin A-Acetate 325 GFP

Recommended use of the chemical and restriction on use

Recommended use*: Vitamin

* The "Recommended use" identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:

BASF CORPORATION
100 Park Avenue
Florham Park, NJ 07932, USA

Telephone: +1 973 245-6000

Emergency telephone number

CHEMTREC: 1-800-424-9300
BASF HOTLINE: 1-800-832-HELP (4357)

Other means of identification

Synonyms: Retinyl acetate

2. Hazards Identification

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Skin Sens.	1	Skin sensitization
Repr.	1B (unborn child)	Reproductive toxicity
Combustible Dust	Combustible Dust (1)	Combustible Dust

Label elements

Pictogram:

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Signal Word:
Danger

Hazard Statement:

H317 May form combustible dust concentration in air.
H317 May cause an allergic skin reaction.
H360 May damage the unborn child.

Precautionary Statements (Prevention):

P280 Wear protective gloves/protective clothing/eye protection/face protection.
P261 Avoid breathing dust/fume/gas/mist/vapours/spray.
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P272 Contaminated work clothing should not be allowed out of the workplace.

Precautionary Statements (Response):

P303 + P352 IF ON SKIN (or hair): Wash with plenty of soap and water.
P308 + P311 IF exposed or concerned: Call a POISON CENTER or doctor/physician.
P362 + P364 Take off contaminated clothing and wash it before reuse.

Precautionary Statements (Storage):

P405 Store locked up.

Precautionary Statements (Disposal):

P501 Dispose of contents/container to hazardous or special waste collection point.

Hazards not otherwise classified

Avoid dust development and deposition - dust explosion risk. Take precautionary measures against static discharges. If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture. The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

3. Composition / Information on Ingredients

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

<u>CAS Number</u>	<u>Weight %</u>	<u>Chemical name</u>
57-50-1	32.0 - 36.0%	Sucrose
9005-25-8	24.0 - 28.0%	starch
127-47-9	11.0 - 14.0%	Vitamin A Acetate
10191-41-0	1.0 - 3.0%	D,L-alpha-Tocopherol

4. First-Aid Measures

Description of first aid measures

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General advice:

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

If on skin:

Immediately wash thoroughly with soap and water, seek medical attention.

If in eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

If swallowed:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11., Further important symptoms and effects are so far not known.

Indication of any immediate medical attention and special treatment needed

Note to physician

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, carbon dioxide, dry powder, foam

Unsuitable extinguishing media for safety reasons:
water jet

Additional information:
Avoid whirling up the material/product because of the danger of dust explosion.

Special hazards arising from the substance or mixture

Hazards during fire-fighting:
carbon oxides

The substances/groups of substances mentioned can be released in case of fire. Burning produces harmful and toxic fumes. Dust explosion hazard.

Advice for fire-fighters

Protective equipment for fire-fighting:
Firefighters should be equipped with self-contained breathing apparatus and turn-out gear.

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Further information:

Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.
Cool endangered containers with water-spray.

Dusty conditions may ignite explosively in the presence of an ignition source causing flash fire.

6. Accidental release measures

Further accidental release measures:

Avoid dispersal of dust in the air (i.e., clearing dust surfaces with compressed air). Avoid the formation and build-up of dust - danger of dust explosion. Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition.

Personal precautions, protective equipment and emergency procedures

Avoid dust formation. Do not breathe dust. Use breathing apparatus if exposed to vapours/dust/aerosol. Ensure adequate ventilation. Use personal protective clothing. Information regarding personal protective measures see, section 8.

Environmental precautions

Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For small amounts: Contain with dust binding material and dispose of.

For large amounts: Sweep/shovel up.

Dispose of absorbed material in accordance with regulations. Avoid raising dust. Cleaning operations should be carried out only while wearing breathing apparatus.

Nonsparking tools should be used.

7. Handling and Storage

Precautions for safe handling

Avoid dust formation. Processing machines must be fitted with local exhaust ventilation.

Protection against fire and explosion:

Avoid dust formation. Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition. Routine housekeeping should be instituted to ensure that dusts do not accumulate on surfaces. Dry powders can build static electricity charges when subjected to the friction of transfer and mixing operations. Provide adequate precautions, such as electrical grounding and bonding, or inert atmospheres. Refer to NFPA 654, Standard for the Prevention of Fire and Dust Explosions from the Manufacturing, Processing, and Handling of Combustible Particulate Solids (2013 Edition) for safe handling.

Conditions for safe storage, including any incompatibilities

No applicable information available.

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place. Protect from air. Protect from the effects of light.

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

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Sucrose	OSHA PEL	PEL 15 mg/m3 Total dust ; PEL 5 mg/m3 Respirable fraction ; TWA value 15 mg/m3 Total dust ; TWA value 5 mg/m3 Respirable fraction ;
	ACGIH TLV	TWA value 10 mg/m3 ;
starch	OSHA PEL	PEL 15 mg/m3 Total dust ; PEL 5 mg/m3 Respirable fraction ; TWA value 15 mg/m3 Total dust ; TWA value 5 mg/m3 Respirable fraction ;
	ACGIH TLV	TWA value 10 mg/m3 ;

Advice on system design:

It is recommended that all dust control equipment such as local exhaust ventilation and material transport systems involved in handling of this product contain explosion relief vents or an explosion suppression system or an oxygen deficient environment. Ensure that dust-handling systems (such as exhaust ducts, dust collectors, vessels, and processing equipment) are designed in a manner to prevent the escape of dust into the work area (i.e., there is no leakage from the equipment). Use only appropriately classified electrical equipment and powered industrial trucks.

Personal protective equipment

Respiratory protection:

Breathing protection if dusts are formed.

Wear a NIOSH-certified (or equivalent) particulate respirator.

Hand protection:

Chemical resistant protective gloves, Suitable materials, rubber, plastic

Eye protection:

Wear safety goggles (chemical goggles) if there is potential for airborne dust exposures.

Body protection:

Body protection must be chosen based on level of activity and exposure.

General safety and hygiene measures:

Under no circumstances should the product come into contact with the skin of pregnant women or be inhaled by them. Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Avoid inhalation of dust. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

9. Physical and Chemical Properties

Form:	free flowing, powder	
Odour:	mild	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
Colour:	light yellow	
pH value:	not applicable	
Melting point:	not determined	
<i>Information on: Vitamin A Acetate</i>		
melting range:	58.6 - 59.3 °C	(measured)

Boiling point:	not determined	

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Information on: Vitamin A Acetate

Boiling point:	(1,013 hPa) The substance / product decomposes therefore not determined. Study scientifically not justified.	

:	No applicable information available.	
Flash point:	not applicable, the product is a solid	
Flammability:	not highly flammable	(UN Test N.1 (ready combustible solids))
Flammability of Aerosol Products:	not applicable, the product does not form flammable aerosoles	
Lower explosion limit:	For solids not relevant for classification and labelling.	
Upper explosion limit:	For solids not relevant for classification and labelling.	
Autoignition:	No applicable information available.	
SADT:	> 75 °C Heat accumulation / Dewar 500 ml (SADT, UN-Test H.4, 28.4.4)	
Vapour pressure:	negligible	
Information on: Vitamin A Acetate		
Vapour pressure:	< 0.00001 hPa (20 °C) yes	(measured)

Relative density:	No applicable information available.	
Bulk density:	approx. 600 kg/m ³	
Vapour density:	not applicable	
Partitioning coefficient n-octanol/water (log Pow):	not applicable for mixtures	
Information on: Vitamin A Acetate		
Partitioning coefficient n-octanol/water (log Pow):	9.4 (25 °C)	(measured)

Thermal decomposition:	155 °C	
Viscosity, dynamic:	not applicable, the product is a solid	
Viscosity, kinematic:	No applicable information available.	
Solubility in water:	dispersible	
Solubility (quantitative):	No applicable information available.	
Molar mass:	328.5 g/mol	
Evaporation rate:	negligible	

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:
No data available.

Oxidizing properties:
Based on its structural properties the product is not classified as oxidizing.

Minimum ignition energy:
> 30 - < 100 mJ, Inductivity: 1 mH, Grain size distribution: 10 - 80 µm (VDI 2263, sheet 1, 2.5)
The product is capable of dust explosion.

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Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

Dust explosion hazard.

Conditions to avoid

Avoid dust formation. Avoid all sources of ignition: heat, sparks, open flame. Avoid moisture. Avoid light. avoid atmospheric oxygen

Incompatible materials

oxidizing agents

Hazardous decomposition products

Decomposition products:

Hazardous decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated.

Thermal decomposition:

155 °C

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity

Assessment of acute toxicity: Virtually nontoxic after a single ingestion.

Information on: Vitamin A Acetate

Assessment of acute toxicity: Of low toxicity after single ingestion.

Oral

Information on: Vitamin A Acetate

Type of value: LD50

Species: rat (male/female)

Value: > 2,000 mg/kg (BASF-Test)

No mortality was observed. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Assessment other acute effects

No applicable information available.

Irritation / corrosion

Information on: Vitamin A Acetate

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Assessment of irritating effects: May cause slight irritation to the skin. Not irritating to the eyes.

Sensitization

Assessment of sensitization: Sensitization after skin contact possible.

Information on: D,L-alpha-Tocopherol

Assessment of sensitization:

Caused skin sensitization in animal studies.

Aspiration Hazard

No aspiration hazard expected.

Chronic Toxicity/Effects

Repeated dose toxicity

Assessment of repeated dose toxicity: Based on available Data, the classification criteria are not met.

Information on: Vitamin A Acetate

Assessment of repeated dose toxicity: Repeated exposure to large quantities may affect certain organs.

Genetic toxicity

Assessment of mutagenicity: Based on available Data, the classification criteria are not met.

Information on: Vitamin A Acetate

Assessment of mutagenicity: In the majority of tests performed (bacteria/microorganisms/cell cultures) a mutagenic effect was not found. A mutagenic effect was also not observed in in-vivo assays. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity: Based on available Data, the classification criteria are not met.

Information on: Vitamin A Acetate

Assessment of carcinogenicity: No reliable data was available concerning carcinogenic activity.

Reproductive toxicity

Assessment of reproduction toxicity: Based on available Data, the classification criteria are not met.

Teratogenicity

Assessment of teratogenicity: The substance caused malformations/developmental toxicity in laboratory animals.

Information on: Vitamin A Acetate

Assessment of teratogenicity: May cause harm to the unborn child.

Other Information

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The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

Symptoms of Exposure

The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11., Further important symptoms and effects are so far not known.

12. Ecological Information

Toxicity

Aquatic toxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. At the present state of knowledge, no negative ecological effects are expected.

Aquatic toxicity

Information on: Vitamin A Acetate

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. No toxic effects occur within the range of solubility. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Information on: D,L-alpha-Tocopherol

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish

Information on: Vitamin A Acetate

LC50 (96 h) 1.37 mg/l, *Oncorhynchus mykiss* (OECD Guideline 203)

The statement of the toxic effect relates to the analytically determined concentration. The LC50 is higher than the solubility limit. Tested above maximum solubility. No toxic effects occur within the range of solubility.

Aquatic plants

Information on: Vitamin A Acetate

EC50 (72 h) 0.103 mg/l (biomass), *Scenedesmus subspicatus* (OECD Guideline 201, static)

The details of the toxic effect relate to the nominal concentration. No toxic effects occur within the range of solubility. Tested above maximum solubility.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms

Information on: Vitamin A Acetate

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OECD Guideline 209 aquatic
activated sludge, domestic/EC20 (180 min): > 1,000 mg/l
The details of the toxic effect relate to the nominal concentration. The product has not been tested.
The statement has been derived from substances/products of a similar structure or composition.

Persistence and degradability

Assessment biodegradation and elimination (H2O)

Information on: Vitamin A Acetate

Moderately/partially biodegradable. Not readily biodegradable (by OECD criteria).

Information on: D,L-alpha-Tocopherol

Not readily biodegradable (by OECD criteria). Biodegradable.

Elimination information

Information on: Vitamin A Acetate

42.2 % CO₂ formation relative to the theoretical value (28 d) (OECD 301B; ISO 9439; 92/69/EEC, C.4-C) (aerobic, activated sludge, domestic)

Bioaccumulative potential

Assessment bioaccumulation potential

Information on: Vitamin A Acetate

Significant accumulation in organisms is not to be expected.

Information on: D,L-alpha-Tocopherol

Accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments

Information on: Vitamin A Acetate

No data available.

Adsorption to solid soil phase is expected.

Additional information

Other ecotoxicological advice:

The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components.

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13. Disposal considerations

Waste disposal of substance:

Observe national and local legal requirements.

Container disposal:

Dispose of in accordance with national, state and local regulations.

14. Transport Information

Land transport

USDOT

Not classified as a dangerous good under transport regulations

Sea transport

IMDG

Not classified as a dangerous good under transport regulations

Air transport

IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Federal Regulations

Registration status:

Chemical TSCA, US blocked / not listed

Food TSCA, US released / exempt

EPCRA 311/312 (Hazard categories): Refer to SDS section 2 for GHS hazard classes applicable for this product.

State regulations

State RTK

PA

CAS Number

57-50-1

Chemical name

Sucrose

Safe Drinking Water & Toxic Enforcement Act, CA Prop. 65:

WARNING: This product can expose you to chemicals including RETINOL ACETATE, which is known to the State of California to cause birth defects or other reproductive harm. For more information, go to www.P65Warnings.ca.gov.

NFPA Hazard codes:

Health: 2 Fire: 1 Reactivity: 0 Special:

HMIS III rating

Health: 2^a Flammability: 1 Physical hazard: 0

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16. Other Information

SDS Prepared by:
BASF NA Product Regulations
SDS Prepared on: 2018/08/16

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END OF DATA SHEET